

PAPER_696: Antennae Galaxies (NGC 4038/4039): Active Merger and Shock-Trigged Star Formation

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Abstract

The Antennae Galaxies (NGC 4038/4039) are the closest major galaxy merger, exhibiting spectacular tidal tails, active starburst regions, and X-ray bright shock ionization zones. UQFF modifies the merger gravity and shock-enhanced star formation.

1. Parameters

Parameter	Value
M_{4038}	$10^{11} M_{\odot}$
M_{4039}	$8 \times 10^{10} M_{\odot}$
Separation	7 kpc
SFR (burst)	$20 M_{\odot}/\text{yr}$

2. UQFF Merger Gravity

$$g_{\text{merge}} = \frac{G(M_1 + M_2)}{r^2} (1 + f_{\text{TRZ}}) + \rho_{\text{shock}} V g \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}}$$

3. Dynamical Friction (Chandrasekhar)

$$t_{\text{df}} = \frac{1.17 v^3}{G^2 M_{\text{tot}} \rho_{\text{avg}} \ln \Lambda}$$

4. UQFF Star Formation Rate

$$\text{SFR}_{\text{UQFF}} = \text{SFR}_{\text{burst}} \frac{\rho_{\text{shock}}}{\rho_{\text{UA}}} \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{UA}}} \right)$$

5. Binding Energy

$$E_{\text{bind}} = -\frac{GM_1 M_2}{d_{\text{sep}}}$$

References

- Hibbard et al. (2001), AJ, 122, 2969
- UQFF Framework, Star-Magic Session 174

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